Technical Note 5 | Corporate Intelligence Course

TECHNICAL NOTE 5

INDUSTRIAL & MANUFACTURING INTELLIGENCE

Industry 4.0, Patent Analysis, Supply Chain Intelligence, and the Agentic CI Pipeline

César Moreno Pascual PhD.
Universidad Francisco de Vitoria

1. Market Context: Industry 4.0

Industry 4.0, the Fourth Industrial Revolution, transforms traditional manufacturing through digital technologies. The goal is creating 'smart factories' that are interconnected, data-driven, and capable of autonomous decisions. For the competitive intelligence analyst, Industry 4.0 creates both new data sources (IoT telemetry, digital twins, predictive maintenance logs) and new intelligence questions ('Is a competitor's factory modernisation signalling capacity expansion or product pivot?').

> **CROSS-REFERENCE:** The techniques for detecting these signals connect directly to TN03's Alternative Data and Early Warning Systems. IoT data, satellite imagery of factories, and patent filings are all alternative data sources that feed weak-signal detection.

1.1 Core Technologies

Internet of Things (IoT). Sensors throughout production lines generate real-time operational data: machine health (vibration, temperature, pressure), production monitoring (output tracking, OEE — Overall Equipment Effectiveness), energy management, and environmental compliance. For CI purposes, a competitor's IoT maturity indicates their ability to respond quickly to market changes — a direct input to the Capability dimension of the AMC Framework.

> **EASY EXAMPLE — Fitness Tracker Analogy:** IoT in a factory is like a fitness tracker on your wrist. It continuously monitors heart rate, steps, and sleep. Without it, you only notice problems when you feel ill. With it, you spot trends early — just as factory sensors spot a machine degrading before it breaks.

Digital Twins. Virtual replicas of physical assets for simulation and optimisation. Uses include process simulation (test changes virtually before implementation), predictive modelling, operator training, and remote monitoring. A competitor investing heavily in digital twins signals sophistication in their manufacturing CI capability — they can test strategic responses faster.

Predictive Maintenance. Machine learning models predict equipment failures before they occur: failure prediction, remaining useful life (RUL) estimation, optimal maintenance scheduling, and parts inventory optimisation. This technology directly affects a competitor's cost structure and capacity reliability — both critical inputs to ECOMO evaluation.

1.2 Utilities Sector Integration

Utilities are increasingly interconnected with industrial operations through smart grids (real-time load balancing, demand response), distributed generation (on-site renewables, microgrids), energy storage (peak shaving and resilience), and water management (smart metering, leak detection). For CI, a competitor's utility integration level is a leading indicator of operational efficiency and cost competitiveness.

2. Applying the CI Framework Pathway to Manufacturing

This section maps the complete CI analytical pathway — from phenomenon detection through to strategic decision — to the manufacturing context. Every framework from TN01–TN03 has a specific application in industrial intelligence.

> **KEY INSIGHT:** The CI Framework Pathway for Manufacturing: Phenomenon detected (TN03 Early Warning) → OSINT collection (TN03 Alternative Data) → AMC profiling (TN01) → Scenario generation (TN02 Shell Method) → Exchange Value evaluation (TN01) → ECOMO cost-benefit (TN02) → Strategic decision. Every step maps to a specific agentic node in TN10/TN11.

2.1 AMC Framework in Manufacturing

The Awareness-Motivation-Capability (AMC) Framework from TN01 predicts competitor responses to your strategic moves. In manufacturing, each dimension has industry-specific indicators:

AMC Dimension	Manufacturing Indicators	Evidence Sources
AWARENESS (Will they detect our move?)	CI function sophistication, patent monitoring activity, trade show presence, industry association leadership, supply chain visibility tools	Patent citation patterns, conference participation, LinkedIn CI team size, supplier network overlap
MOTIVATION (Will they care enough to respond?)	Revenue at stake in the product category, factory utilisation rate (idle capacity = more motivated), strategic priority signalled in annual reports, executive compensation tied to market share	Annual reports, earnings calls, capacity utilisation data, executive compensation filings, factory satellite imagery
CAPABILITY (Can they mount an effective response?)	Available CapEx budget, production flexibility (reconfigurable lines vs. dedicated), engineering talent pool, supplier relationships, regulatory approvals already held	SEC filings (CapEx plans), patent portfolio depth, job postings (hiring surge?), facility permits, import records (new equipment orders)

> **CROSS-REFERENCE:** This directly applies TN01 Section 2 (AMC Framework). The scoring methodology (HIGH/MEDIUM/LOW) and the response probability estimation are identical — only the evidence sources change for the manufacturing context.

> **EASY EXAMPLE — Chess Analogy:** AMC in manufacturing is like assessing a chess opponent. Awareness = do they see your move? (Are they watching the board?) Motivation = do they care about this part of the board? (Is their king nearby?) Capability = can they actually counter? (Do they have the right pieces in position?) A competitor with a full patent monitoring team (high A), 60% of revenue in your product category (high M), but depleted CapEx after a recent acquisition (low C) is ‘aware and angry but unable to punch back.’

2.2 Scenario Planning: Shell Method for Manufacturing

The Shell Method from TN02 provides a structured five-step process for manufacturing scenario development:

Step 1 — Identify Driving Forces. In manufacturing, driving forces include: technological disruption (additive manufacturing, autonomous robots), regulatory change (carbon emissions standards, trade tariffs, safety regulations), supply chain restructuring (nearshoring, friendshoring), raw material availability, and workforce transformation (skills gap, automation vs. employment).

Step 2 — Determine Critical Uncertainties. Plot driving forces on the 2×2 matrix (Uncertainty × Impact). For a European auto parts manufacturer, the two highest-uncertainty/highest-impact forces might be: (1) Speed of EV adoption vs. ICE persistence, and (2) EU Carbon Border Adjustment Mechanism stringency.

Step 3 — Construct the Scenario Matrix. The 2×2 yields four scenarios, each a coherent narrative of a plausible future. For example: 'Rapid EV + Strict Carbon' = Green Sprint; 'Rapid EV + Loose Carbon' = Electric Pragmatism; 'Slow EV + Strict Carbon' = Regulatory Squeeze; 'Slow EV + Loose Carbon' = Business as Usual.

Step 4 — Develop Scenario Narratives. Each scenario gets a detailed story: what does the competitive landscape look like? Which competitors thrive? What happens to our supply chain?

Step 5 — Identify Signposts. Observable events that indicate which scenario is materialising. For manufacturing: regulatory announcements, competitor patent filings in specific IPC codes, raw material price movements, factory construction starts detected via satellite.

> **CROSS-REFERENCE:** This directly applies TN02 Section 3 (Shell Method). The five-step process and signpost monitoring are identical. The scenario matrix is the same 2×2 tool — only the driving forces are manufacturing-specific.

2.3 War Gaming in Manufacturing

Business war gaming from TN02 Section 1–2 adapts powerfully to manufacturing competitive dynamics. A manufacturing war game typically involves:

Blue Team (Your Company): Proposes a strategic move — e.g., launching a new production line, entering a geographic market, or acquiring a supplier.

Red Team (Key Competitor): Plays the competitor, using AMC-informed intelligence to craft the most aggressive plausible response.

Control Team: Injects market shocks (supply chain disruption, regulatory change, commodity price spike) and adjudicates outcomes.

> **CASE EXAMPLE — Siemens vs. Rockwell Automation:** Imagine Siemens is considering a major push into the North American discrete manufacturing automation market — Rockwell's home turf. A war game would have the Red Team (playing Rockwell) respond with: (1) aggressive pricing on existing contracts, (2) accelerated product development in areas where Siemens is weak, (3) lobbying for 'Buy American'

procurement preferences. The Blue Team must then decide: does the original strategy survive competitive retaliation? What modifications improve resilience across scenarios?

CROSS-REFERENCE: This applies TN02 Sections 1–2 (War Gaming). The Red Team/Blue Team/Control structure is identical. The debrief phase produces the highest-value insights — exactly as described in TN02's 'Debrief Depth' principle.

2.4 Manufacturing Blind Spots

TN03's Blind Spot Analysis is especially critical in manufacturing, where four common blind spots recur:

Misjudging Industry Boundaries. Traditional manufacturers often define competition too narrowly. A steel producer competing with other steel producers may miss the threat from advanced composites or 3D-printed metal parts. Industry 4.0 blurs boundaries further — software companies (e.g., Siemens with Xcelerator, PTC with ThingWorx) increasingly compete in 'manufacturing' via digital platforms.

Overconfidence in Physical Moats. Manufacturers with large installed bases and capital-intensive facilities often assume these assets protect them. But asset-light competitors using contract manufacturing (like many consumer electronics brands) can enter markets without building factories.

Supply Chain Tunnel Vision. Focusing only on Tier 1 suppliers while ignoring Tier 2/3 dependencies. The semiconductor shortage of 2020–2023 revealed how a single sub-component can halt entire production lines.

Technology Substitution Blindness. Failure to monitor technologies that could make your product category obsolete. Kodak's failure to see digital photography is the classic case; in manufacturing, examples include 3D printing substituting injection moulding for low-volume parts.

Countermeasures from TN03 — Devil's Advocacy, Dialectical Inquiry, and Pre-Mortem Analysis — apply directly. A manufacturing Pre-Mortem asks: 'It is 2028 and our new production line has failed. Why?'

> **CROSS-REFERENCE:** This applies TN03 Section 1 (Blind Spots) and Section 1.3 (Countermeasures). The four types of blind spots and three countermeasures map directly to manufacturing-specific manifestations.

3. Patent Intelligence

Patents provide early signals of competitor R&D direction — often 18–24 months before product launch. Patent intelligence is one of the most reliable leading indicators in manufacturing CI.

3.1 Key Patent Indicators

Indicator	What It Reveals	CI Application
Filing Velocity	Year-over-year change in applications by technology area	Acceleration signals increased R&D investment; deceleration may indicate pivot or budget cuts
Technology Classification (IPC/CPC)	Specific technical focus areas	Reveals which problems a competitor is trying to solve; compare with your own R&D roadmap
Citation Analysis	Forward citations indicate technology importance	Highly-cited patents signal foundational technologies that competitors must work around
Inventor Networks	Key talent and collaboration patterns	Identifies critical personnel; changes in inventor teams signal strategic shifts
Geographic Strategy	Filing locations (US, EU, China, etc.)	Reveals market entry plans 18+ months ahead — you only patent where you plan to sell

> **EASY EXAMPLE — Recipe Book Analogy:** Patents are like a chef publishing a new recipe book. The recipes they choose to patent (file) tell you what dishes (products) they plan to serve. The countries where they file tell you which restaurants (markets) they plan to open. If they suddenly start patenting lots of vegan recipes, they are pivoting to plant-based — even if their current menu is all meat.

3.2 Data Sources

USPTO PatentsView API. Free access to US patent applications and grants. Provides full text, images, and structured data. Accessible via REST API for programmatic analysis. Applications are published 18 months after filing.

EPO Espacenet / OPS API. European patent data with global coverage. Includes EP applications plus PCT national phase entries.

> **CROSS-REFERENCE:** Patent databases are OSINT sources from TN03 Section 2. In the agentic pipeline (TN10/TN11), the OSINT Collector agent calls the USPTO PatentsView API as a tool, retrieving patent data that feeds the AMC Analyst and Scenario Planner agents.

4. Supply Chain Intelligence

Import/export data and supply chain mapping reveal production strategies, sourcing dependencies, and upcoming product launches. This is one of the richest alternative data sources for manufacturing CI.

4.1 What Trade Data Reveals

Intelligence Question	Trade Data Signal	Data Source
Where does a competitor source components?	Bills of lading show shipper-consignee relationships	Panjiva (S&P Global), ImportGenius
Are they expanding production?	Increase in equipment/machinery imports	Customs data, Panjiva
Are they launching a new product?	New component types appearing in imports	Bill of lading product descriptions
Are they reshoring/nearshoring?	Geographic shift in supplier origins	Trade flow trend analysis
What is their capacity utilisation?	Raw material import volume vs. historical baseline	Import volume time-series

4.2 Facility Intelligence

Physical asset monitoring through alternative data adds another layer: satellite imagery (construction progress, parking lot activity, environmental compliance), utility filings (power connection requests indicate expansion), permit applications (building and environmental permits), and job postings (hiring patterns reveal capacity plans and technology direction).

> **CROSS-REFERENCE:** Satellite imagery and job postings are alternative data sources from TN03 Section 2. The monitoring frequency matrix from TN03 Section 3.4 guides how often to check each source: satellite imagery monthly, job postings weekly, permit filings monthly.

5. Key Intelligence Indicators & Early Warning System

Applying TN03's Early Warning System framework to manufacturing intelligence requires defining Key Intelligence Topics (KITs), selecting leading and lagging indicators, and establishing alert thresholds.

5.1 Manufacturing KITs

KIT	Leading Indicators	Lagging Indicators	Monitoring Frequency
Competitor Capacity Expansion	Building permits, equipment imports, job postings for operators	Production volume increases, market share shift	Monthly
Technology Disruption	Patent filing velocity in new IPC codes, startup funding rounds	Product launches, market adoption rates	Weekly
Supply Chain Reconfiguration	New supplier relationships in trade data, geographic shift	Cost structure changes in financial filings	Monthly
Regulatory Impact	Draft legislation, industry lobbying spend	Final regulations, compliance costs reported	Quarterly
Pricing Aggression	Competitor bid patterns, distributor channel checks	Market price indices, margin compression	Real-time / Daily

5.2 Alert System Design

Apply TN03's Monitoring Frequency Matrix to manufacturing signals. High volatility + high impact signals (e.g., a competitor's surprise patent filing in your core technology) go to the Critical Alert quadrant: real-time/daily monitoring with immediate escalation. Low volatility + low impact signals (e.g., industry association membership changes) go to the Background Scan quadrant: quarterly review.

> **KEY INSIGHT:** The most dangerous signals in manufacturing CI are the ones that cross industry boundaries — a software company filing hardware patents, a materials company entering your product category. These are the blind spots from Section 2.4 that your Early Warning System must specifically watch for.

6. Integrated Economic Evaluation: Predictive Maintenance CI System

This section applies both the Exchange Value methodology (TN01) and the ECOMO model (TN02) to evaluate a predictive maintenance system investment. This is the complete CI evaluation pathway: from scenario generation through economic quantification.

6.1 Exchange Value Analysis

Following TN01's Exchange Value methodology: Exchange Value = Reference Value + Differential Value.

Reference Value (Do Nothing). Continue with reactive maintenance. Current annual cost of unplanned downtime: €500,000. Equipment replacement costs on current lifecycle: €200,000/year. Spare parts inventory carrying cost: €50,000/year. Total annual cost of status quo: €750,000.

Differential Value (Predictive Maintenance). The change in economic outcomes vs. the reference scenario. Reduced unplanned downtime (80% reduction): +€400,000. Extended equipment life (15% longer): +€100,000. Optimised scheduling: +€50,000. Reduced spare parts inventory: +€30,000. Total annual differential benefit: +€580,000.

> **CROSS-REFERENCE:** This applies TN01 Section 4 (Exchange Value). The reference scenario establishes the baseline; differential value measures the improvement. The methodology is identical to the competitive scenario evaluation in TN01 — only the domain changes.

6.2 ECOMO Cost-Benefit Model

Development Costs (One-Time): Sensor installation €200,000 + Software licensing €300,000 + System integration €100,000 + Training €50,000 = €650,000 total.

Annual Operating Costs: Data processing €50,000 + Software maintenance €30,000 + Dedicated staff (0.5 FTE) €40,000 + Sensor calibration €20,000 = €140,000/year.

Net Annual Benefit: €580,000 − €140,000 = €440,000.

6.3 Scenario-Weighted NPV

Using TN02's scenario-weighted expected value calculation, we evaluate the investment across three scenarios generated by our Shell Method analysis:

Scenario	Probability	NPV (10%, 5yr)	Expected Value
A: Full Adoption — predictive maintenance delivers maximum value	50%	€1,018,000	€509,000
B: Partial Adoption — integration challenges reduce benefits by 40%	35%	€393,000	€137,550
C: Technology Failure — system underperforms, salvage value only	15%	−€450,000	−€67,500

Scenario	Probability	NPV (10%, 5yr)	Expected Value
EXPECTED VALUE	100%		€579,050

The positive expected value of €579,050 across all three scenarios indicates this is a robust investment even under uncertainty. Note how the scenario-weighted approach from TN02 provides a more rigorous assessment than a single-point NPV estimate.

> **CROSS-REFERENCE:** This applies TN02 Section 5 (ECOMO model) with scenario weighting from Section 5.3. The cost categories (Development, Operating, Opportunity) and value categories (Monetary, Psychological) are directly from the ECOMO framework.

> **EASY EXAMPLE — Holiday Insurance Analogy:** Scenario-weighted NPV is like deciding whether to buy travel insurance. Scenario A (great holiday, 70%): you lose the insurance premium but have a fantastic trip. Scenario B (minor issue, 20%): insurance covers a cancelled flight. Scenario C (major disruption, 10%): insurance covers hospital bills. The expected value across all scenarios tells you whether the insurance is worth the premium — just as our three scenarios tell us whether the predictive maintenance investment is worth €650,000.

7. Agentic CI Architecture for Manufacturing Intelligence

This section maps the 10-stage agentic CI pipeline from TN10/TN11 to the manufacturing and industrial context. Every agent node in the pipeline has a specific manufacturing application.

> **KEY INSIGHT:** The technology serves the methodology, not the other way around. The agentic pipeline automates the same CI Framework Pathway you apply manually in Sections 2–6 — AMC profiling, scenario generation, Exchange Value calculation, ECOMO evaluation — but at machine speed with built-in anti-hallucination safeguards.

7.1 The 10-Stage Pipeline Applied to Manufacturing

Stage	Agent	Manufacturing Application	TN Reference
1. Trigger	Early Warning Monitor	Detects anomaly: competitor patent filing spike in additive manufacturing IPC codes	TN03 Sec 3
2. Collect	OSINT Collector	Scrapes USPTO PatentsView API, Panjiva trade data, satellite imagery feeds, job posting aggregators	TN03 Sec 2
3. Profile (AMC)	AMC Analyst	Scores competitor on Awareness (CI function), Motivation (revenue at stake), Capability (CapEx, talent)	TN01 Sec 2
4. Predict	Response Predictor	Estimates probability and timeline of competitor response to our factory expansion	TN01 Sec 2.4
5. Scenarios	Scenario Generator (Shell)	Builds 2×2 scenario matrix using driving forces (EV adoption speed × carbon regulation)	TN02 Sec 3
6. Exchange Value	EV Calculator	Computes Reference Value (do nothing) + Differential Value per scenario branch	TN01 Sec 4
7. ECOMO	ECOMO Evaluator	Calculates NPV at 10% over 5 years; scenario-weighted expected value	TN02 Sec 5
8. Verify	Source Verifier	Cross-checks every claim against source documents; flags unsourced assertions	TN11 Sec 2
9. Bias Check	Bias Detector	Checks for manufacturing blind spots: industry boundary myopia, physical moat overconfidence	TN03 Sec 1
10. Report	CI Reporter	Generates executive brief with Go/No-Go recommendation, confidence scores, full citation chain	TN01 Sec 5

7.2 Manufacturing-Specific Tool Use

In the agentic architecture from TN10, Tool Use is one of Andrew Ng's four design patterns. For manufacturing CI, the agent's tool belt includes:

Tool (API)	Data Retrieved	Pipeline Stage
USPTO PatentsView API	Patent applications, grants, citations, inventor networks by IPC code	2. Collect
Panjiva / ImportGenius	Import/export records, bill of lading data, supplier relationships	2. Collect
Satellite Imagery API (Planet Labs, Orbital Insight)	Factory activity indices, construction monitoring, parking lot counts	2. Collect
SEC EDGAR	10-K filings with CapEx plans, risk factors, segment revenue	2. Collect + 3. AMC
Job Posting Aggregators (LinkedIn, Indeed)	Hiring patterns, skills demanded, location of new roles	2. Collect + 3. AMC
ChromaDB Vector Store	Historical CI analyses, past AMC scores, previous scenario outcomes	All stages (RAG)

EASY EXAMPLE — Detective's Toolkit: Each tool is like a different instrument in a detective's toolkit. The patent API is like checking criminal records (past behaviour). Trade data is like tracking someone's purchases (what they're buying reveals intent). Satellite imagery is like staking out a location (physical activity). The detective (agent) picks the right tool for each question — and ChromaDB is the detective's case file, storing everything for future reference.

7.3 RAG for Manufacturing CI

Retrieval-Augmented Generation (TN10) ensures the manufacturing CI agent grounds its analysis in actual data rather than hallucinating. The ChromaDB vector store holds four collections adapted for manufacturing:

Collection	Contents	Update Frequency
competitor_intel	Patent filings, trade data, facility observations, job postings per competitor	Weekly
market_trends	Industry reports, commodity prices, regulatory announcements, technology forecasts	Monthly
internal_strategy	Our own R&D roadmap, production plans, cost structure, strategic priorities	Quarterly
historical_archive	Past CI analyses, previous AMC scores, scenario outcomes, war game transcripts	After each analysis

CROSS-REFERENCE: RAG is covered in TN10 (Section on RAG and Embeddings). The three-step process — INDEX documents as embeddings, RETRIEVE top-k similar chunks, GENERATE grounded answer — is identical. Only the document types change for the manufacturing vertical.

7.4 Anti-Hallucination in Manufacturing CI

Manufacturing CI has specific hallucination risks that the TN11 anti-hallucination framework must address:

Hallucination Risk	Manufacturing Example	Countermeasure (TN11)
Fabricated data	Agent invents a patent number or import volume that does not exist	Source Verification: every claim must cite a retrievable URL or document ID
Outdated information	Agent uses 3-year-old capacity data when factory has since expanded	RAG with time-stamped documents + freshness checks
False correlation	Agent claims patent filing spike 'proves' a product launch when filings are for a different division	Confidence Scoring: HIGH/MEDIUM/LOW/UNCERTAIN framework
Overconfident AMC scores	Agent scores Capability as 5/5 based on a single news article	AMC Consistency Check: score must align with evidence volume and quality

CROSS-REFERENCE: Anti-hallucination patterns are from TN11 Sections 2–4. The confidence scoring framework (HIGH = 2+ authoritative sources, MEDIUM = 1 reliable source, LOW = inference only, UNCERTAIN = conflicting data) applies unchanged to manufacturing intelligence.

8. Practical Exercise: MegaRetail vs. ValueMart — Smart Factory War Game

This exercise integrates all frameworks covered in this note. You will apply the complete CI pathway to a manufacturing competitive scenario.

8.1 Scenario Brief

MegaRetail, a large consumer goods manufacturer, has detected (via patent monitoring) that ValueMart, a low-cost competitor, has filed 47 patents in the last 6 months related to automated warehouse fulfilment and robotic packaging. ValueMart's import records show a 300% increase in robotic equipment from Japan and South Korea. Satellite imagery reveals construction activity at a greenfield site near their existing factory.

8.2 Your Tasks

Task 1 — AMC Analysis (TN01): Score ValueMart on all three dimensions using the evidence provided. Estimate their response probability and timeline if MegaRetail were to announce a competing smart factory initiative.

Task 2 — Scenario Planning (TN02): Identify two critical uncertainties for MegaRetail. Construct a 2×2 scenario matrix and develop brief narratives for each quadrant. Define at least two signposts per scenario.

Task 3 — Exchange Value + ECOMO (TN01/TN02): Calculate the Exchange Value of MegaRetail's smart factory investment using a reference scenario (continue with current operations) and differential value across your scenarios. Apply ECOMO to compute scenario-weighted expected value.

Task 4 — Blind Spots (TN03): Identify at least two blind spots that MegaRetail's leadership might have. Propose specific countermeasures (Devil's Advocacy, Pre-Mortem, or Dialectical Inquiry).

Task 5 — Agentic Design (TN10/TN11): Design a CLAUDE.md file for a Claude Code project that would automate this analysis. Specify: which OSINT tools the Collector agent would call, what the AMC Analyst's system prompt should include, and where human-in-the-loop checkpoints are needed.

9. Discussion Questions

1. Design a patent monitoring system for tracking competitor R&D in autonomous vehicles. Which IPC codes would you monitor? How would you connect patent velocity changes to AMC scoring?

2. How would you use import/export data to detect a competitor's manufacturing capacity expansion? What are the leading indicators vs. lagging confirmation signals?

3. A competitor has filed 15 patents in additive manufacturing (3D printing) for aerospace components. Apply the full CI Framework Pathway: Early Warning → OSINT → AMC → Scenarios → Exchange Value → ECOMO → Decision. What is your recommendation?

4. Design a manufacturing war game for a European steel producer facing competition from a Chinese state-subsidised entrant. What roles would you assign? What shock events would the Control Team inject?

5. How would the 10-stage agentic CI pipeline from TN10/TN11 handle a scenario where the OSINT Collector finds conflicting data — satellite imagery shows factory activity increasing but import records show raw material orders declining? Trace the pipeline's response through the Verify and Bias Check stages.

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Appendix: CI Framework Cross-Reference Map

This table maps every framework used in this technical note to its source TN, its section within this document, and its corresponding agentic pipeline stage.

Framework	Source TN	TN05 Section	Agentic Stage (TN10/TN11)
AMC Framework	TN01 Sec 2	2.1	Stage 3: Profile → AMC Analyst agent
Exchange Value	TN01 Sec 4	6.1	Stage 6: Value → EV Calculator
Shell Method (Scenario Planning)	TN02 Sec 3	2.2	Stage 5: Scenarios → Scenario Generator
War Gaming	TN02 Sec 1–2	2.3	Multi-agent debate (TN11 Ex.5)
ECOMO Model	TN02 Sec 5	6.2–6.3	Stage 7: ECOMO → ECOMO Evaluator
Blind Spots	TN03 Sec 1	2.4	Stage 9: Bias Check → Bias Detector
Alternative Data	TN03 Sec 2	3–4	Stage 2: Collect → OSINT Collector
Early Warning Systems	TN03 Sec 3	5	Stage 1: Trigger → Early Warning Monitor
RAG + ChromaDB	TN10	7.3	All stages (vector retrieval)
Anti-Hallucination	TN11 Sec 2–4	7.4	Stage 8: Verify → Source Verifier
LangGraph Pipeline	TN11 Sec 3	7.1	Full orchestration with human-in-the-loop
Confidence Scoring	TN11 Sec 2.4	7.4	Stage 8: Verify → Confidence Scorer